

TECHNICAL REPORT 002

Project SIMILARITY: Learning Routed Timing Laws for Architecture Search

Causal topology intervention, held-out prediction, and route-robust two-candidate selection

RESEARCH LANE

CAUSAL TIMING LAW / ROUTE-ROBUST SELECTION

REPORT ID

KRN / TR-002

DATE

21 September 2026

STATUS

Final-route research report

AUTHOR

Sushanth Dasari

ABSTRACT

Routing topology is an architecture variable

Project SIMILARITY tests whether operand-distribution topology creates a size-dependent final-routed timing effect that can be learned, predicted out of sample, and used by an architecture-selection policy. The corrected causal study completed **96/96** ECP5 routes with exact one-DSP-per-PE mapping. A frozen differential law learned only from square arrays predicts broadcast timing tax on unseen rectangular arrays and disjoint route seeds with **0.3039 ns** mean absolute error, **0.3815 ns** RMSE, and **0.8921** correlation.

96 / 96

corrected causal routes

0.3039 ns

held-out tax MAE

36 / 36

unseen cases beat broadcast

1.44%

mean routed-oracle regret

Confirmed portfolio result

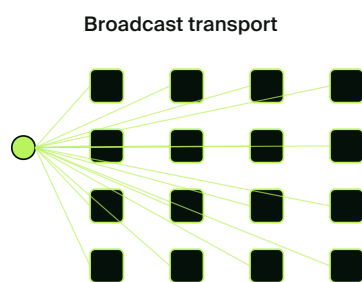
A prospective two-candidate local-transport portfolio completed **72/72** deployment routes and delivered **23.56%** mean routed-latency improvement over the best broadcast-only architecture. The second-ranked candidate was selected in **21/36** cases.

01 / DESIGN

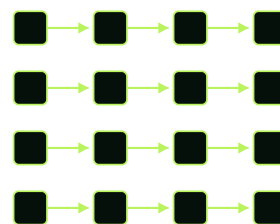
Question and causal intervention

The central question is whether nonlocal operand broadcast causes a routed timing penalty relative to registered nearest-neighbor transport as the processing-element fabric grows. Arithmetic density is held fixed so the intervention targets operand-distribution topology rather than compute count.

Operand-distribution topology as an architecture variable



Registered local transport



Matched arithmetic fabric isolates operand-distribution topology as the intervention.

Figure 1. Causal intervention. Direct broadcast and registered local transport use matched arithmetic fabrics, isolating transport topology as the changed variable.

Leakage-free routed-law evaluation

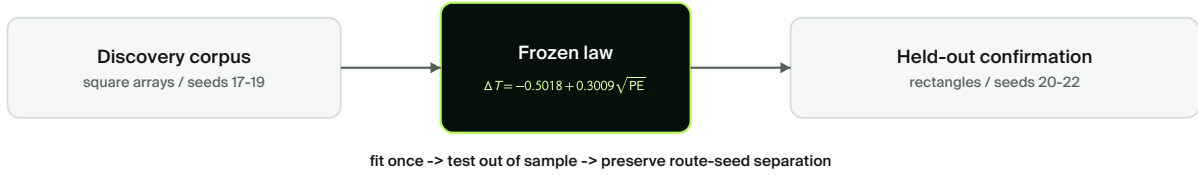


Figure 2. Discovery and held-out separation. The law is fit on square arrays and tested on unseen rectangular geometries and disjoint route seeds.

02 / MODEL

Frozen final-route timing law

Timing is sourced only from authoritative nextpnr post-route report JSON. The discovery model is frozen before held-out results are evaluated.

Frozen law

$$T = \alpha_{\text{device}} + \beta_b \sqrt{\text{PE}} + \text{LOCAL}(\delta_0 + \delta_1 \sqrt{\text{PE}})$$

$$\beta_b = 0.450257, \quad \delta_0 = 0.501805, \quad \delta_1 = -0.300915$$

$$\text{broadcast slope} = 0.450257 \text{ ns}/\sqrt{\text{PE}}, \quad \text{local slope} = 0.149343 \text{ ns}/\sqrt{\text{PE}}$$

$$\widehat{\Delta T} = -0.501805 + 0.300915 \sqrt{\text{PE}} \text{ ns}$$

Frozen differential timing law on unseen routed pairs

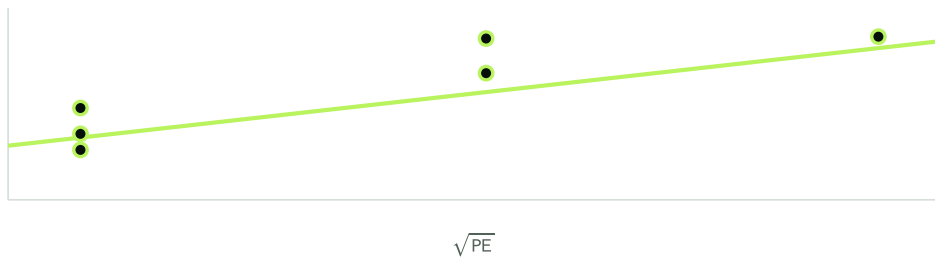
MAE = 0.3039 ns / $r = 0.8921$ 

Figure 3. Frozen differential law on six unseen routed geometry/device pairs. The prediction tracks the observed broadcast-minus-local timing tax with 0.3039 ns MAE and 0.8921 correlation.

03 / UNSEEN GEOMETRIES

Held-out confirmation

Device	Geometry	Broadcast	Local	Observed tax	Predicted tax	Local gain
25K	3x7	10.332 ns	9.629 ns	0.703 ns	0.877 ns	6.80%
45K	3x7	10.941	9.648	1.293	0.877	11.82%
45K	5x9	11.774	9.990	1.784	1.517	15.15%
85K	3x7	10.752	9.823	0.929	0.877	8.64%
85K	5x9	11.855	9.583	2.272	1.517	19.16%
85K	7x11	12.719	10.421	2.298	2.139	18.07%

6 / 6

positive broadcast tax

0.3815 ns

held-out RMSE

0.8921

prediction correlation

13 / 13

causal gates passed

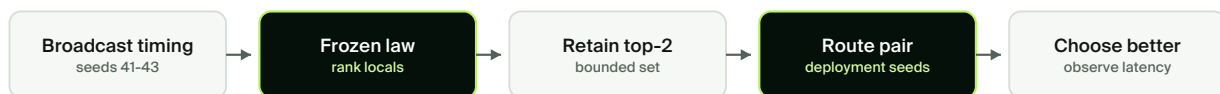
Median route-seed coefficient of variation remains below **2.21%** in every split/topology combination, well inside the frozen **6%** stability ceiling.

04 / SELECTION POLICY

Prospective deployment refinement

The first prospective deployment study already produced strong average gains from a single frozen local choice. Its route-seed sensitivity motivated a stronger decision rule: retain two candidates rather than changing any confirmation threshold.

Two-candidate route-robust selection protocol



36/36 beat broadcast / 23.56% mean improvement / 1.44% oracle regret / rank-2 wins 21/36

Figure 4. Two-candidate route-robust selection. Broadcast-only characterization and the frozen timing law produce a bounded local candidate set before deployment routes are observed.

72 / 72

deployment routes complete

23.56%

mean improvement

1.44%

mean oracle regret

21 / 36

rank-2 candidate selected

05 / PORTFOLIO RESULT

Resource trade-off and confirmation

Across the independent portfolio corpus, local transport uses on average 70.57% more flip-flops, 0.00% additional block RAM, and the same DSP population. This exchanges sequential logic for shorter routed critical paths and route-robust deployment choices.

Supported route-robust selection claim

Within the tested ECP5 INT8 GEMM family, a two-candidate portfolio constructed from broadcast-only multi-seed characterization and a frozen causal topology law converts bounded local physical feedback into route-seed-robust workload choices. Across 36 unseen deployment cases it delivered **23.56%** mean latency improvement over the best broadcast-only architecture at **1.44%** mean regret to the full routed oracle.

All 13 portfolio confirmation criteria passed, including route success, exact DSP mapping, two-candidate integrity, positive architecture-pair taxes, non-worse deployment performance on every workload/seed, and explicit resource reporting.

06 / ARTIFACTS

Reproduction map

Artifact	Repository path / identifier
Causal frozen plan	docs/SIMILARITY_ROUTED_LAW_PLAN.md
Causal raw routes	results/similarity_routed_law_combined.csv
Causal summary	results/similarity_routed_law_summary.json
Portfolio frozen plan	docs/SIMILARITY_PORTFOLIO_CONFIRMATION_PLAN.md
Portfolio raw routes	results/similarity_portfolio_combined.csv
Portfolio selections	results/similarity_portfolio_selections.csv
Portfolio summary	results/similarity_portfolio_summary.json

Publication record

Kernellum Research Technical Report 002. The next evidence layer extends the final-route result into K2 physical measurement and broader transfer studies.